



GitHub Copilot 최신 업데이트



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AI Go-To-Market Lead | GitHub Copilot
GitHub Inc.

Agenda

- 01 GitHub Copilot : Your AI Team
- 02 GitHub Copilot Updates
- 03 GitHub Internal AI-Driven Development
- 04 GitHub Certifications



GitHub Copilot: Your AI Team





AI-powered developer platform



INTEGRATIONS & APIs



COPILOT



COLLABORATION



PRODUCTIVITY

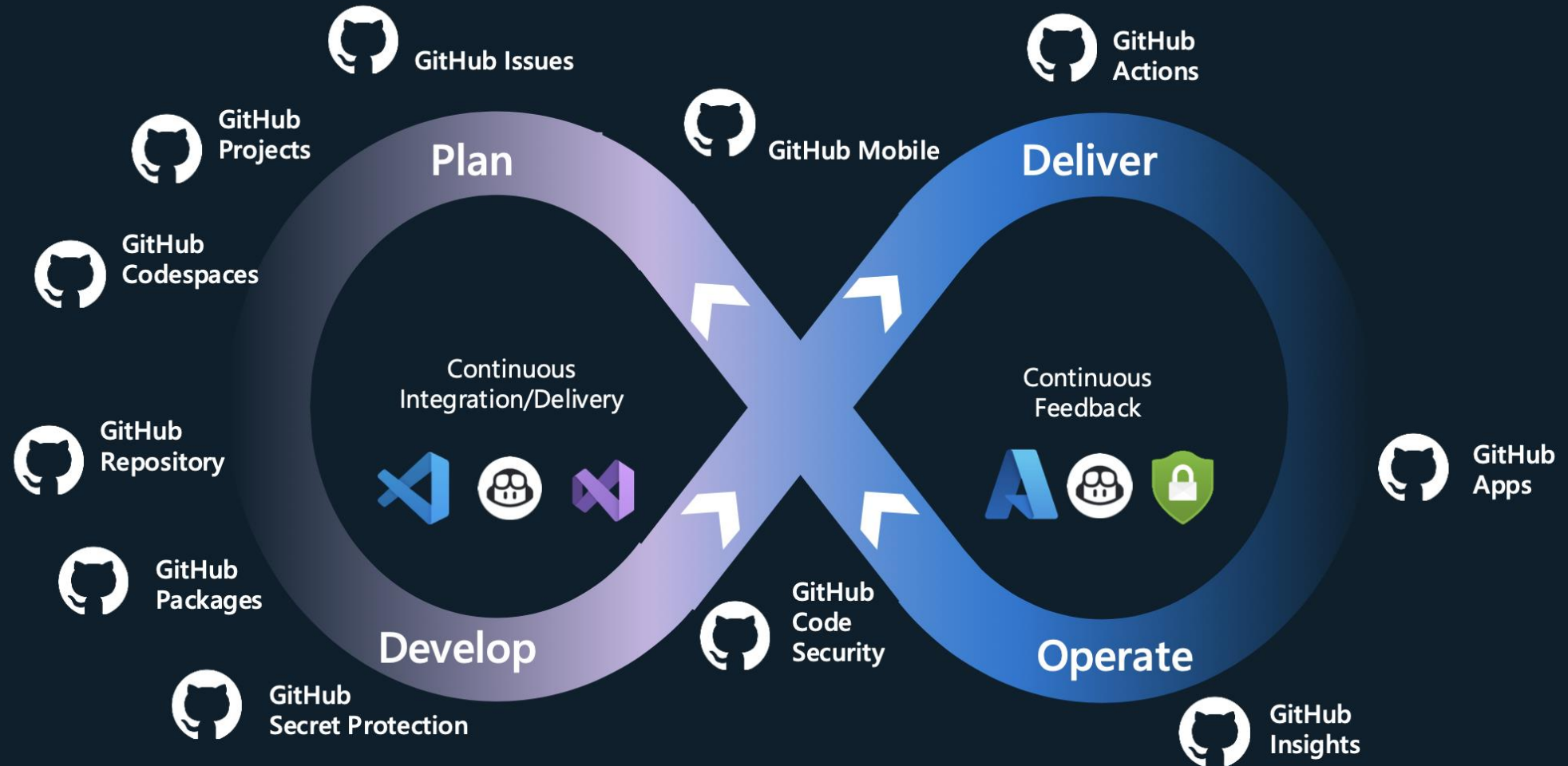


SECURITY



SCALE

GitHubによるSDLCのモダナイゼーション



GitHub and the

Plan

Develop

Delivery

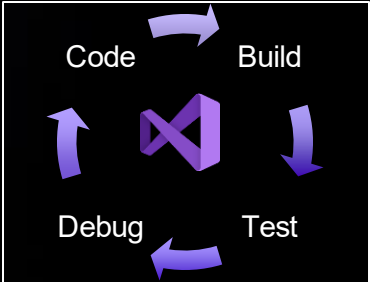
Operate

Project Management
Kanban Board

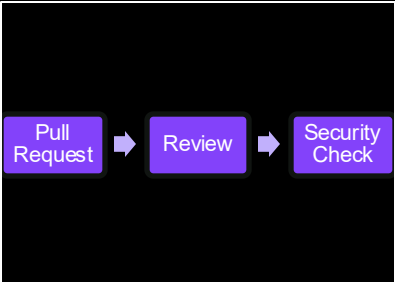


Project
Owner

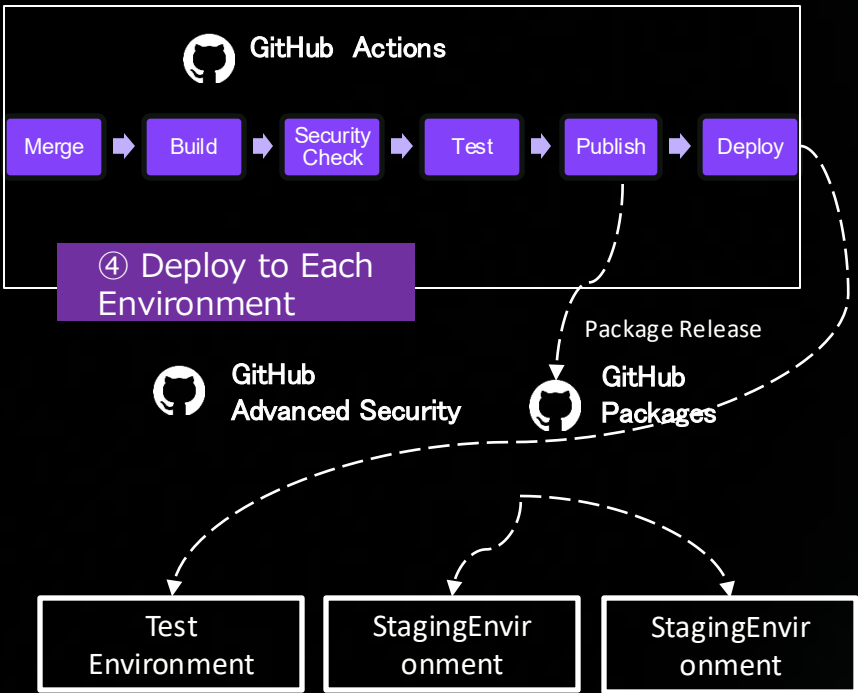
Set Up Dev Environment
Code Development



CodeReview
Source Code Analysis



Test / Build / Release



① Building the
Development
Environment

GitHub Codespaces



DeveloperA



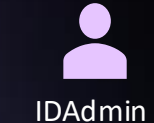
Partner / Contractor
(MFA Required)



GitHub Copilot

③ AI Review and
Security Check

② AI-Assisted
Development Support

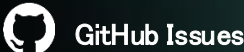


IDAdmin

Identity / Au



Coding Standards, Base
Code



GitHub Issues

Bug Ticket Creation



GitHub
Discussions

Q&A and Discussion



AI Security
Vulnerability Triage,
Remediation



GitHub Copilot

Existing Code Explanation
and Summary



GitHub Repository

CURRENT SCENARIO

Platform activity is surging



✓
14B

There were 1 billion commits in 2025.

Now, it's 275M per week, on pace for 14B this year, if growth remains linear.

SPOILER: IT WON'T.

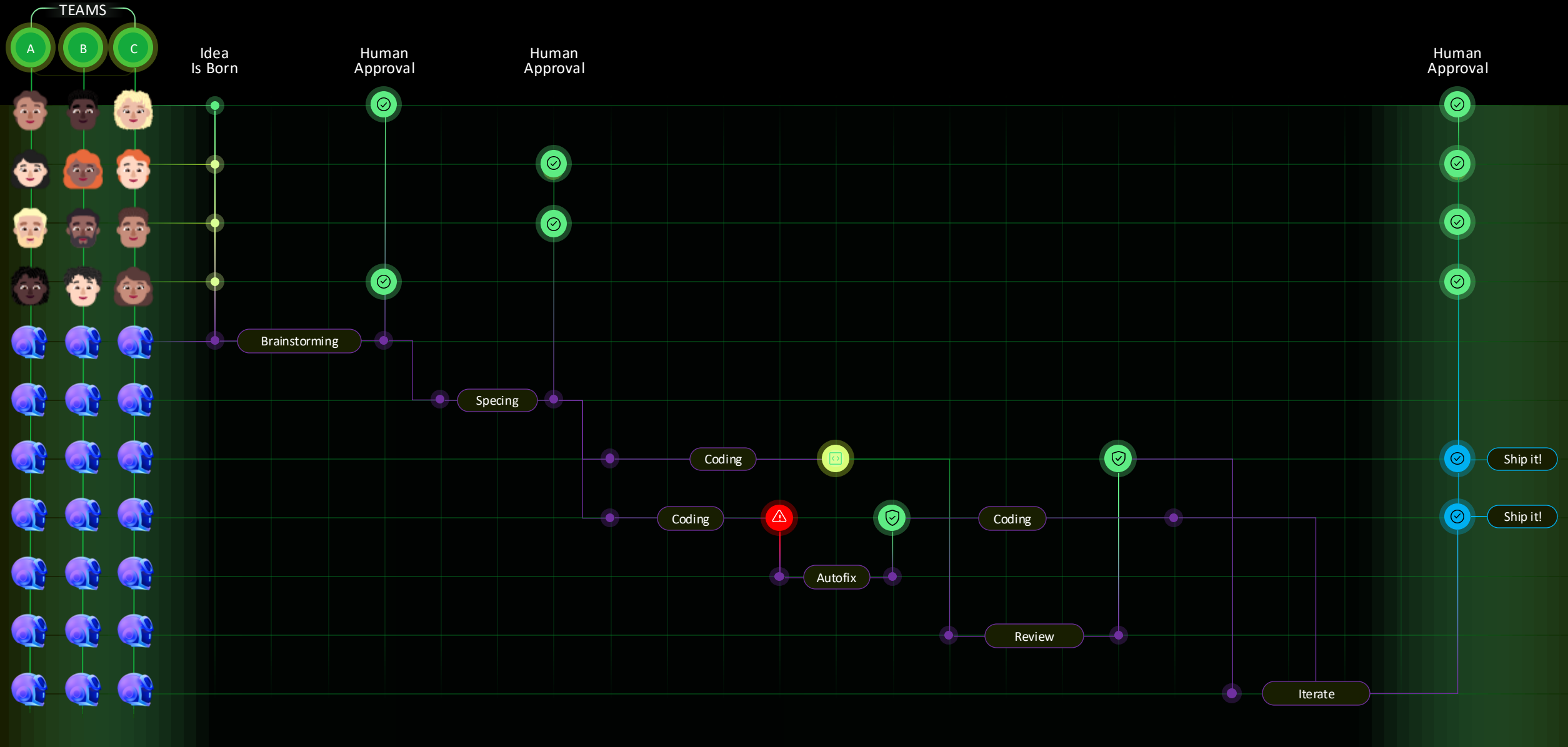
Github Actions has grown from 500M minutes/week in 2023 to 1B minutes/week in 2025, **and now 2.1B/week this week so far.**



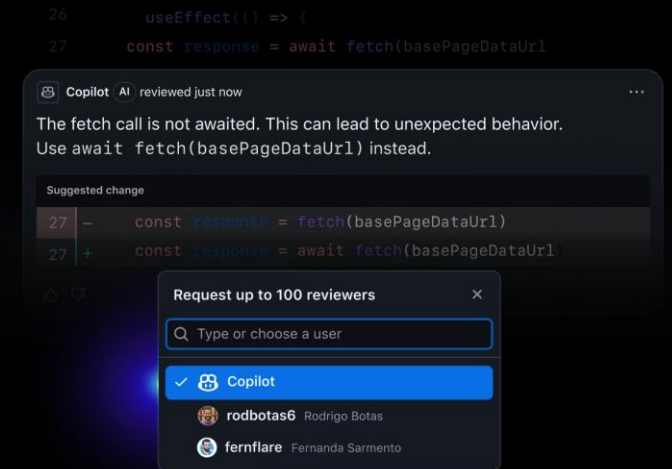
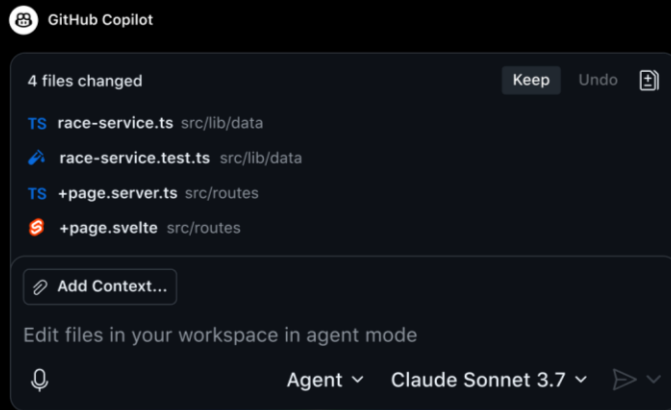
2.1B minutes/week



Agent-native software development



GitHub Copilot : Your AI Team



Agent Mode + CLI

Core Features

- Synchronous & Interactive Workflows
- Multi-file edits across the codebase VS Code real-time debugging

Best For

- Complex tasks requiring immediate feedback
- Multi-step work with human oversight
- Debugging and testing interactions

Coding agent + Agentic Workflow

Core Features

- Asynchronous & Autonomous Development
- Runs in the cloud or your own data center, creating pull requests

Best For

- Well-defined feature additions and bug fixes
- GitHub Issues assigned as tasks
- Background processing while you focus on other work

Code review

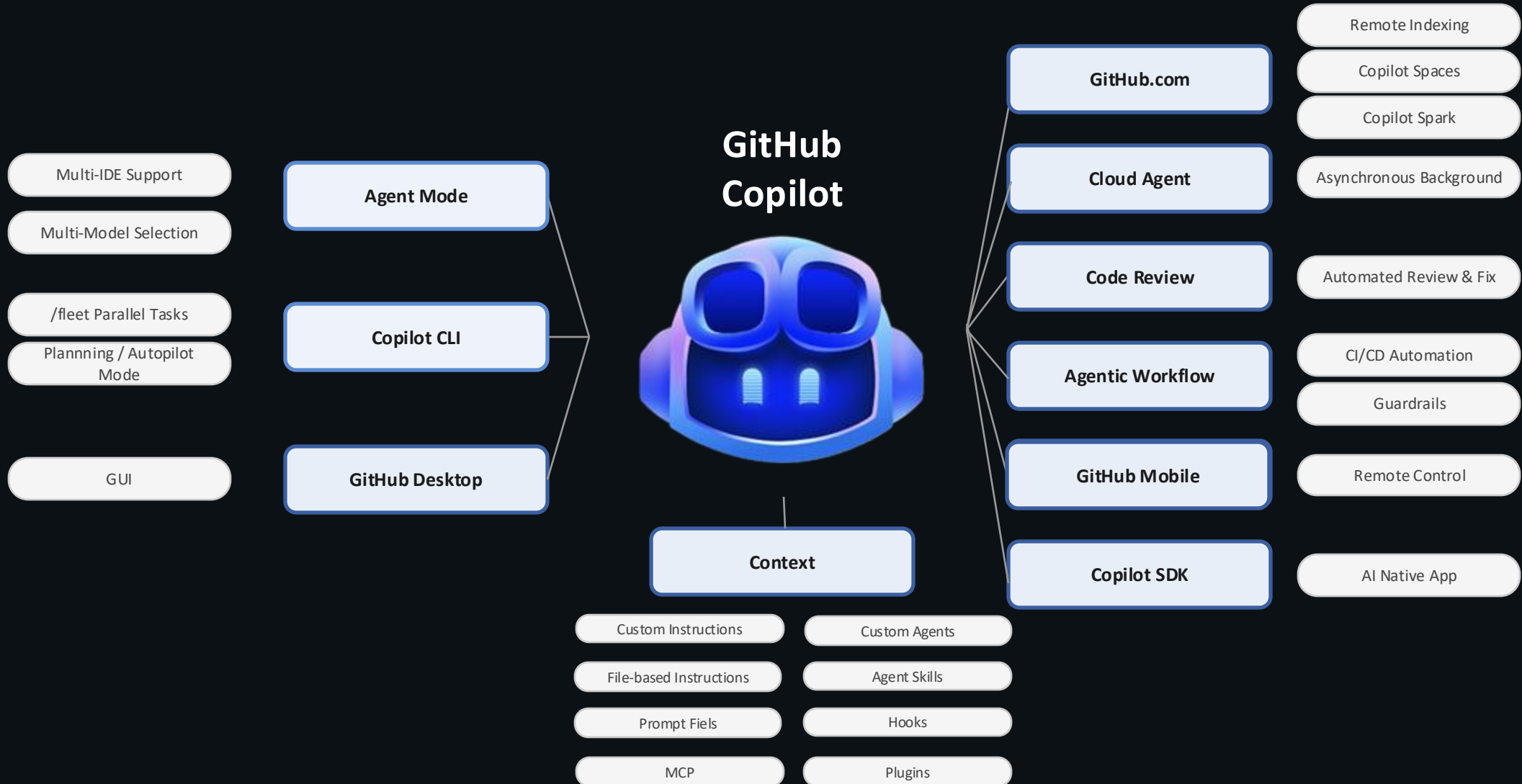
Core Features

- Context-aware PR reviews
- Customizable to match your coding standards

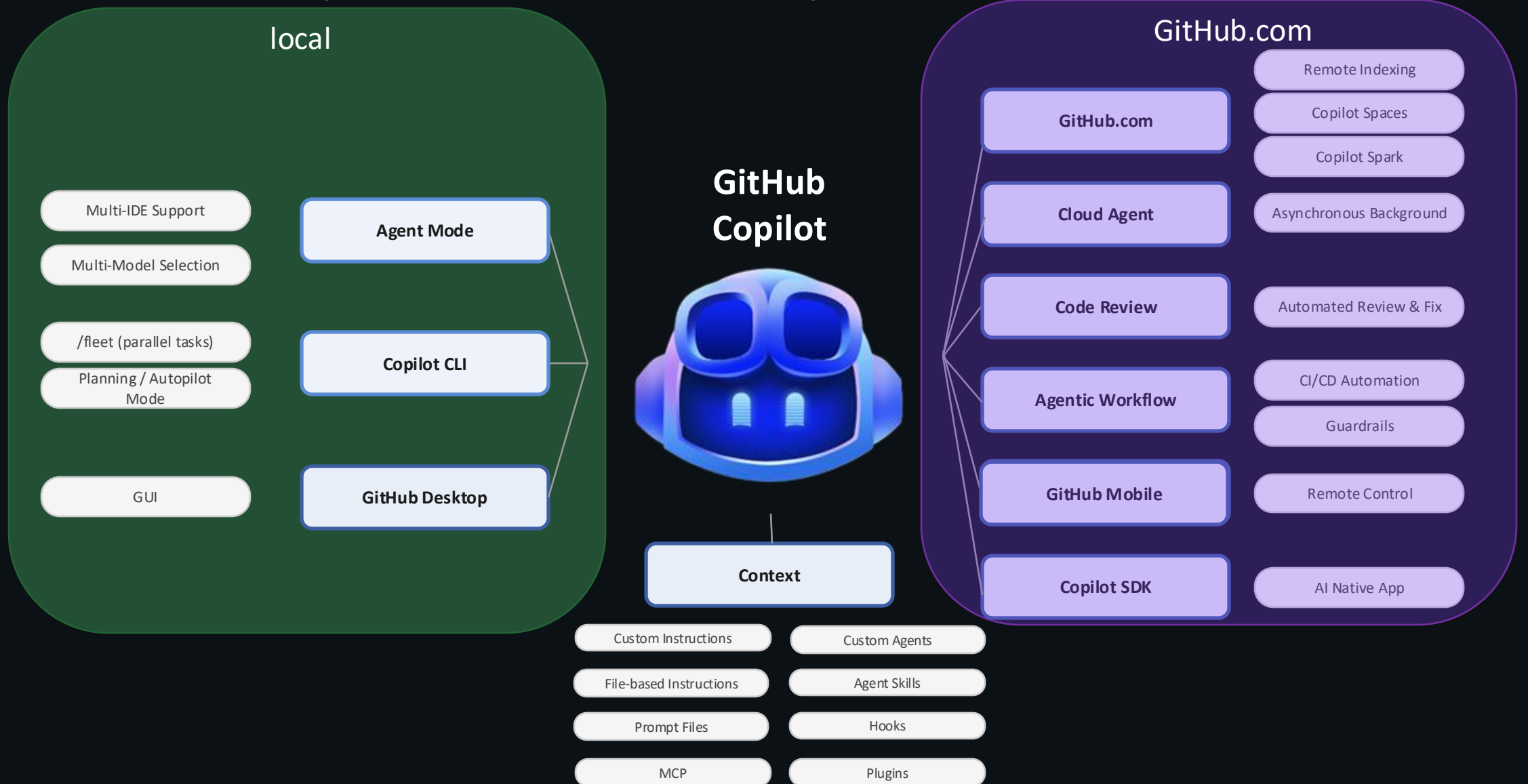
Best For

- Reduces the burden of manual reviews and fixes
- AI accelerates development — detect risks early, maintain quality

GitHub Copilot Feature Map



GitHub Copilot Feature Map



GitHub Updates

1. GitHub Copilot CLI
2. GitHub Copilot SDK
3. Agentic Workflow



GitHub Copilot CLI



```
GitHub Copilot x + v
PS D:\Projects\github\terminal-copilot-sdk> copilot

GitHub Copilot v0.0.400
Describe a task to get started.

Pick a model with /model. Delegate changes with an AI-generated PR using /delegate.
Enter ? to see all commands. Copilot uses AI, so always check for mistakes.

• No copilot instructions found. Run /init to generate a copilot-instructions.md file for this project.

D:\Projects\github\terminal-copilot-sdk[. main*] claude-opus-4.5 (3x)

> Type @ to mention files or / for commands

shift+tab cycle node
```



Copilot CLI is growing quickly

174

Active
Pull Requests

106

Merged
Pull Requests

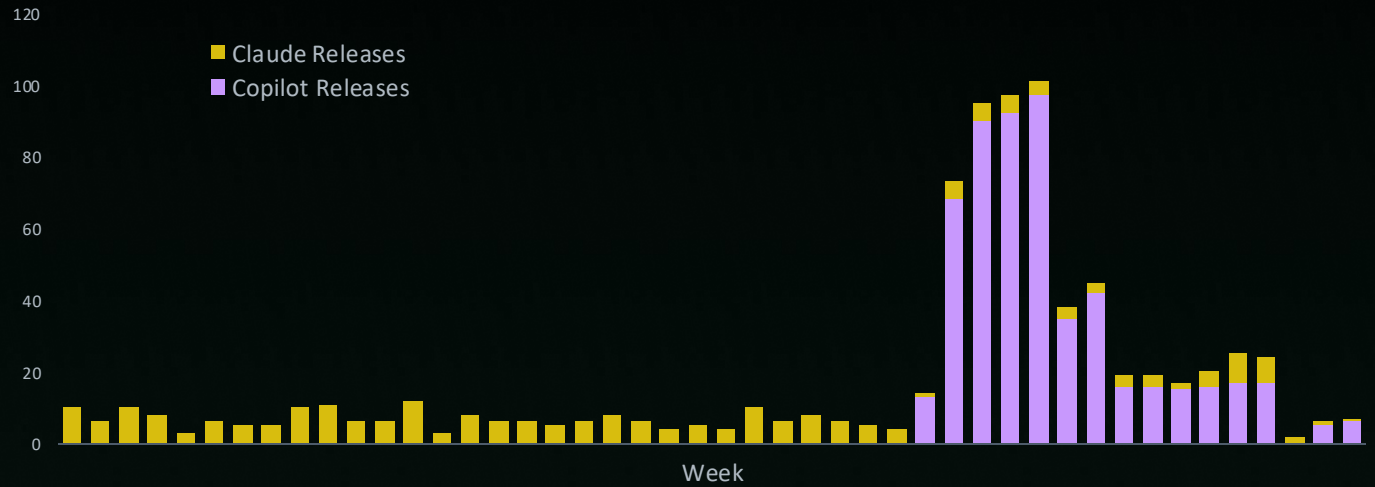
68

Open
Pull Requests

9,678

Commits

Weekly Releases by Product
Copilot CLI vs Claude



Copilot CLI (sonnet-4.5)

Claude Code (sonnet-4.5)

SWE Bench

352/500 (70%)

358/500 (71.6%)

Terminal Bench

48/95 (50.3%)

48/95 (50.5%)

```
skills — GitHub Copilot — copilot - node /opt/homebrew/bin/copilot --yolo — 119x36
[patrick@FC7Q0F77YL skills % copilot --yolo]

 GitHub Copilot v0.0.400
Describe a task to get started.

Pick a model with /model. Delegate changes with an AI-generated PR using /delegate.
Enter ? to see all commands. Copilot uses AI, so always check for mistakes.

• 💡 No copilot instructions found. Run /init to generate a copilot-instructions.md file for this project.

~/projects/skills[⌘ feature*] claude-opus-4.5 (3x)

> /add-dir

  /add-dir <directory>      Add a directory to the allowed list for file access
  /agent                    Browse and select from available agents (if any)
  /allow-all, /yolo        Enable all permissions (tools, paths, and URLs)
  /clear, /new              Clear the conversation history
  /compact                  Summarize conversation history to reduce context window usage
  /context                  Show context window token usage and visualization
  /cwd, /cd [directory]    Change working directory or show current directory
  /delegate [prompt]       Delegate changes to remote repository with AI-generated PR
  /diff                     Review the changes made in the current directory
  /exit, /quit              Exit the CLI
```

Overview

Copilot CLI delivers AI-powered coding assistance directly in your terminal, enabling developers to stay in their workflow.* GA on February 25, 2026

Key Features

- Real-Time Assistant
- Multi-File Context Awareness
- Code Generation & Editing
- Tool Execution & Shell Commands
- IDE-Agnostic Experience
- Headless Mode
- Mode Switching as Needed
- Agent Skills & MCP Server Integration

GitHub Copilot CLI – Mode

 planning



Shift+TabPress Shift+Tab to switch to plan mode and collaborate with Copilot on planning.

 autopilot



Shift+TabPress Shift+Tab to switch to autopilot mode and let Copilot autonomously complete complex tasks.

 /fleet



Deploy a fleet to parallelize any task and orchestrate many tasks at once.

Mode Change	How to Use
Shift + Tab	Planning Mode/ Autopilot Mode Switching
Command	/fleet





Powerful Built-in Agents

Four built-in agents designed for performance and versatility.



Explore

grep/glob/view
Fast codebase exploration using grep/glob/view. Supports parallel execution



Task

Runs commands like tests, builds, linters, and formatters. Provides concise completion summaries.



General

CLIFull-featured agent using CLI tools. Handles complex multi-step tasks.



Review

Reviews code changes. Detects only bugs, security vulnerabilities, and logic errors.



Personalized Custom Agents

Create .agent.md files in .github/agents/ or ~/.github/agents/ to use them.

```
---
name: Display Name
description: What the agent does (required)
tools: ["*"] # Optional - defaults to all tools
mcp-servers: # Optional
  server-name:
    command: npx
    args: ["@some/mcp-server"]
---
# Agent Instructions
```

Frontmatter Field Reference

Field	Required	Details
name	No	Display name (defaults to filename)
description	Yes	Description of when to invoke the agent
tools	No	Array of available tools (default: all)
mcp-servers	No	MCP server configuration
infer	No	Whether the agent can be auto-invoked (default: true)

*See custom agents [docs](#) for details.

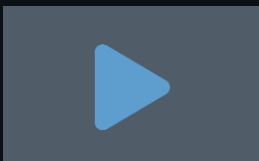
Demo



A terminal window with a title bar that reads "copilot-metrics-dashboard-eng — zsh — 103x30". The terminal content shows a shell prompt "williamzhang@J2T0Q7HF2D" followed by the command "copilot-metrics-dashboard-eng % co" and a cursor.

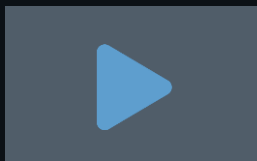
```
williamzhang@J2T0Q7HF2D copilot-metrics-dashboard-eng % co
```

Everything you need **for development.**



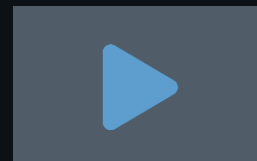
Code Generation

Use natural language to iteratively improve codebases of any size or instantly bring new ideas to life.



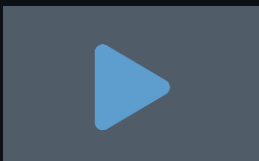
Quality Review

Integrate automated reviews into your workflow to reduce team burden while maintaining quality.



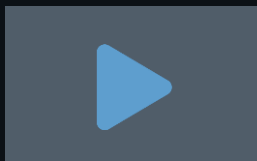
Automation

Helps configure agent automation and software development lifecycle automation.



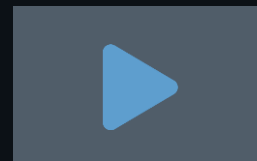
Repository & Environment Understanding

Investigate, plan, and implement code without opening files



Scheduled Execution

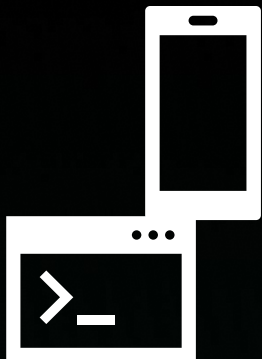
Run in headless mode to automatically process tasks at scheduled times.



Workflow Asset Creation

Leverage Skills, Custom Agents, and Memory to record your work style and turn it into reusable task assets





GitHub Copilot CLI Remote Control

```
urlist | 23 | [icons]

[stash all your changes and get back to a clean main] X

• Todo dependency 11 links added
  extend-group-types → read-next-docs
  update-group-data-access → extend-group-types
  update-group-apis → update-group-data-access
  update-group-drafts → extend-group-types
  build-grouped-dnd-ui → update-group-drafts
  update-public-group-rendering → update-group-data-access
  wire-compose-payloads → update-group-apis
  wire-compose-payloads → update-group-drafts
  add-group-tests → wire-compose-payloads
  add-group-tests → update-public-group-rendering
  validate-grouped-lists → add-group-tests

• Plan ready for review – Add named groups embedded on list records and... · summary: "- Add named groups em...
  ↳ Plan approved! Exited plan mode.

• Plan approved. I'll wait for your next prompt before implementing.

~/dev/burkeholland/urlist [feature/grouped-lists]
> █

5 background /sessions · / commands · ? help GPT-5.5 · low(24%)

[nav icons] feature/grouped-lists [status icons]
```




GitHub Copilot SDK



TypeScript



Python



.NET



Go



Rust



Java



Demo

```
...lan Copilot Metrics Dashboard — workiq • node /opt/homebrew/bin/copilot ...M_PROGRAM=Apple_Terminal SHELL=/bin/zsh TERM=xterm-256color +
```

Tests

- 24 tests passing (vittest): NDJSON parser, cache TTL/expiry, all transformer functions

To run

```
# Set your GitHub PAT in .env.local
echo "GITHUB_TOKEN=ghp_..." >> .env.local
npm run dev      # Start dev server
npm test        # Run tests
```

Required token scopes: `manage_billing:copilot` or `read:enterprise` + `read:org`

- Connected to Visual Studio Code
- Total usage est: 12 Premium requests
- API time spent: 11m 54s
- Total session time: 3h 16m 15s
- Total code changes: +2629 -130
- Breakdown by AI model:
 - Claude Opus 4.6 (1M context)(Internal only) 4.7m in, 52.4k out, 4.5m cached (Est. 12 Premium requests)

~/Documents/Temp/copilot-metrics-dashboard-eng [./main*%] Claude Opus 4.6 (6x)

> Describe a plan. Type @ to mention files, # for issues/PRs, / for commands, or ? for shortcuts

plan • shift+tab switch mode • ctrl+y view/edit plan @.env.local

GitHub Copilot SDK (Public Preview)

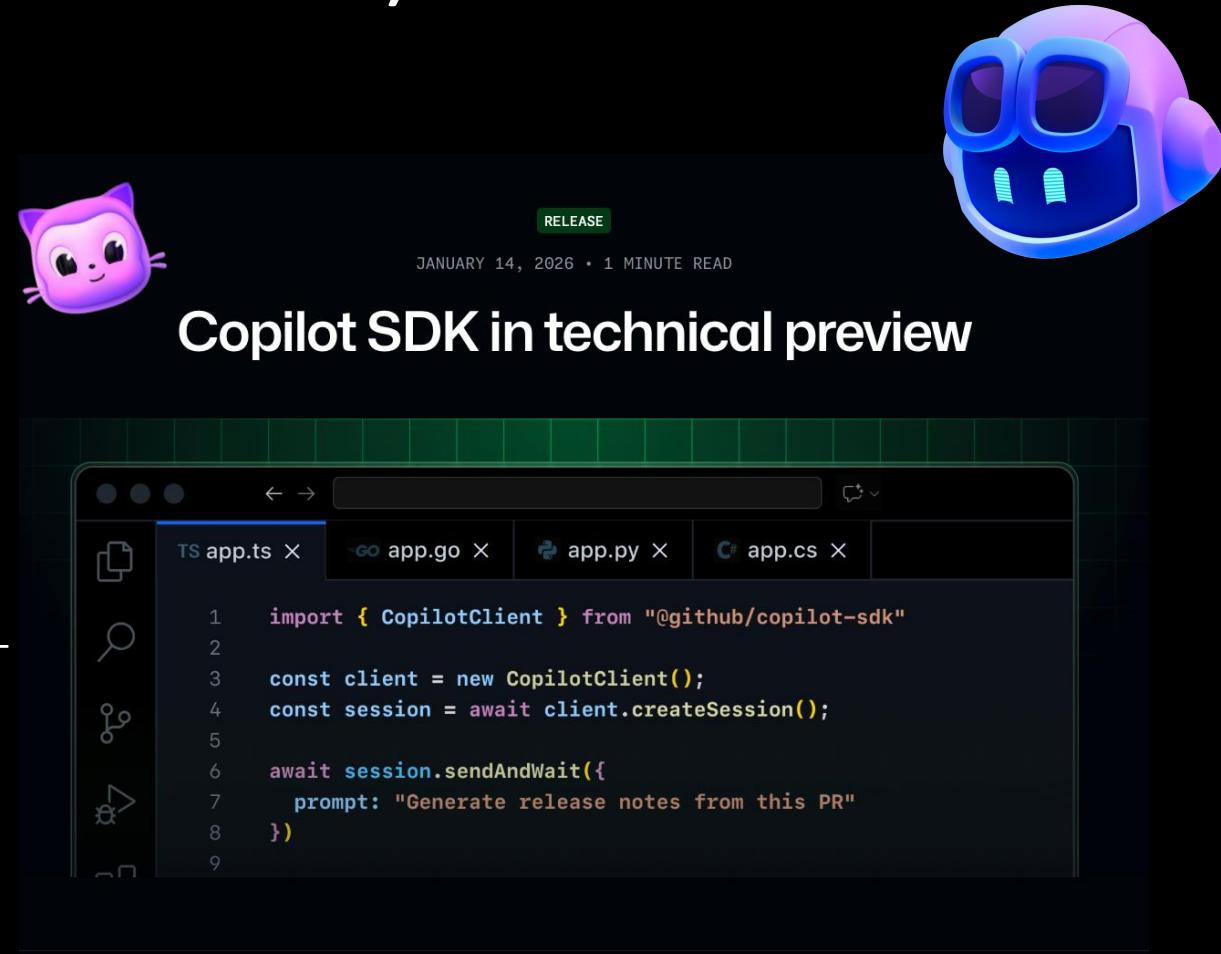
An SDK for developers who want to embed GitHub Copilot CLI into their apps and scripts SDK

Supported Languages

- 6 languages: TypeScript / Python / Go / .NET / Rust / Java

Key Features

- **Multi-turn conversations** - Maintains session history for context-aware dialogue
- **Tool execution** - Define custom tools that can be invoked during conversations
- **Lifecycle management** - Control clients and sessions programmatically

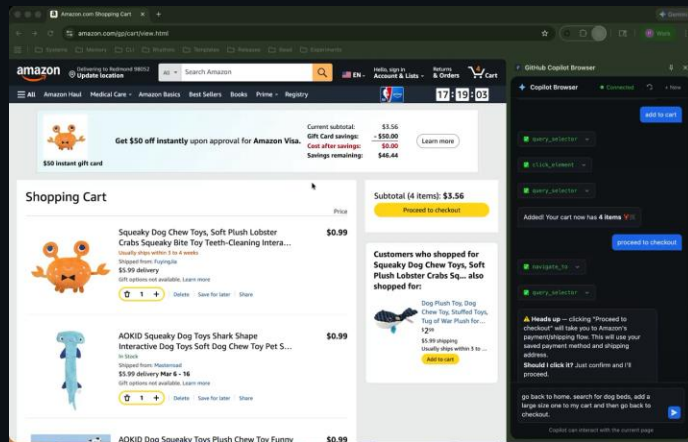


Demo

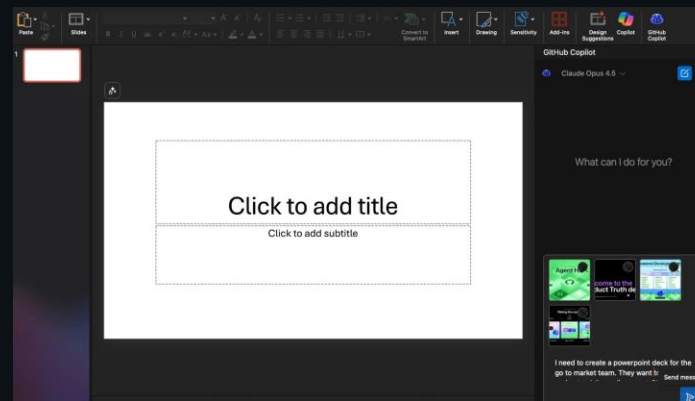
The image shows a VS Code editor window with the following components:

- Explorer (Left):** Displays the project structure for `copilot-metrics-dashboard-eng`. The `src` directory is expanded, showing `components`, `hooks`, and `lib`. The `lib` directory is further expanded, showing files like `cache.ts`, `chat-tools.ts`, `config.ts`, `copilot-chat.ts` (selected), `github-api.ts`, `ndjson-parser.ts`, `tool-logger.ts`, `transformers.ts`, and `types.ts`.
- Editor (Center):** Displays the content of `src > lib > TS copilot-chat.ts > resolveCopilotCliPath`. The code includes imports from `@github/copilot-sdk` and `@/lib/tool-logger`, and a function `resolveCopilotCliPath` that checks for the `COPILOT_CLI_PATH` environment variable and falls back to the local node_modules bin directory.
- Terminal (Bottom):** Shows the command prompt for the user `williamzhang@J2T0Q7HF2D` in the `copilot-metrics-dashboard-eng` directory.
- Status Bar (Bottom):** Indicates the current file is `main*`, the update is ready, and the cursor is at `Ln 14, Col 48`.

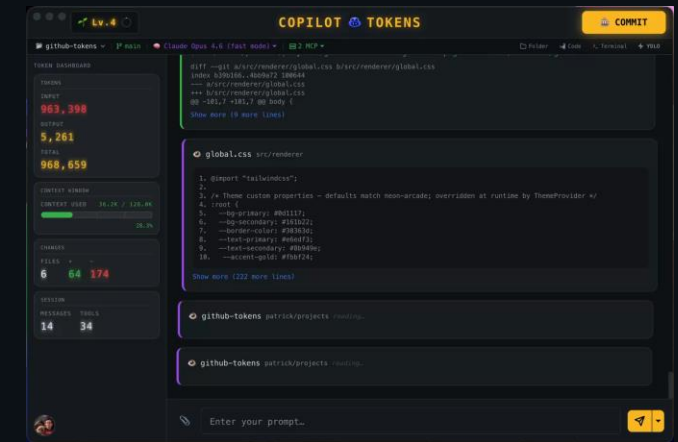
App Integration



Browser Plugins
Integrate into your browser to automate interactions



Office Plugins
Embed Copilot in Office apps for advanced operations



Chat Interface
A Copilot chat web app that can be shared with team members



GitHub Agentic Workflow



GitHub Agentic Workflow (Technical Preview)

Overview

- AI agents automate tasks on GitHub Actions
- Currently in technical preview (announced February 13, 2026)

What It Can Do

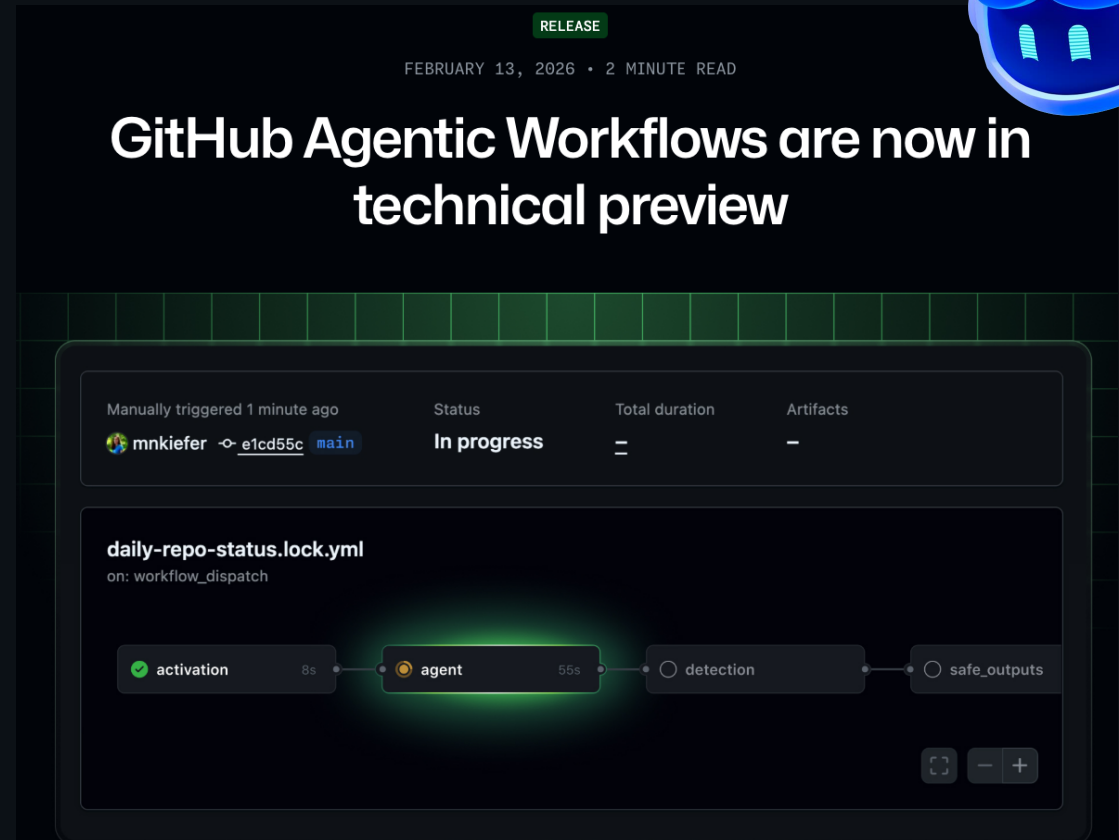
- Automatic issue triage
- PR reviews
- CI error analysis
- Documentation management and more

Key Benefits

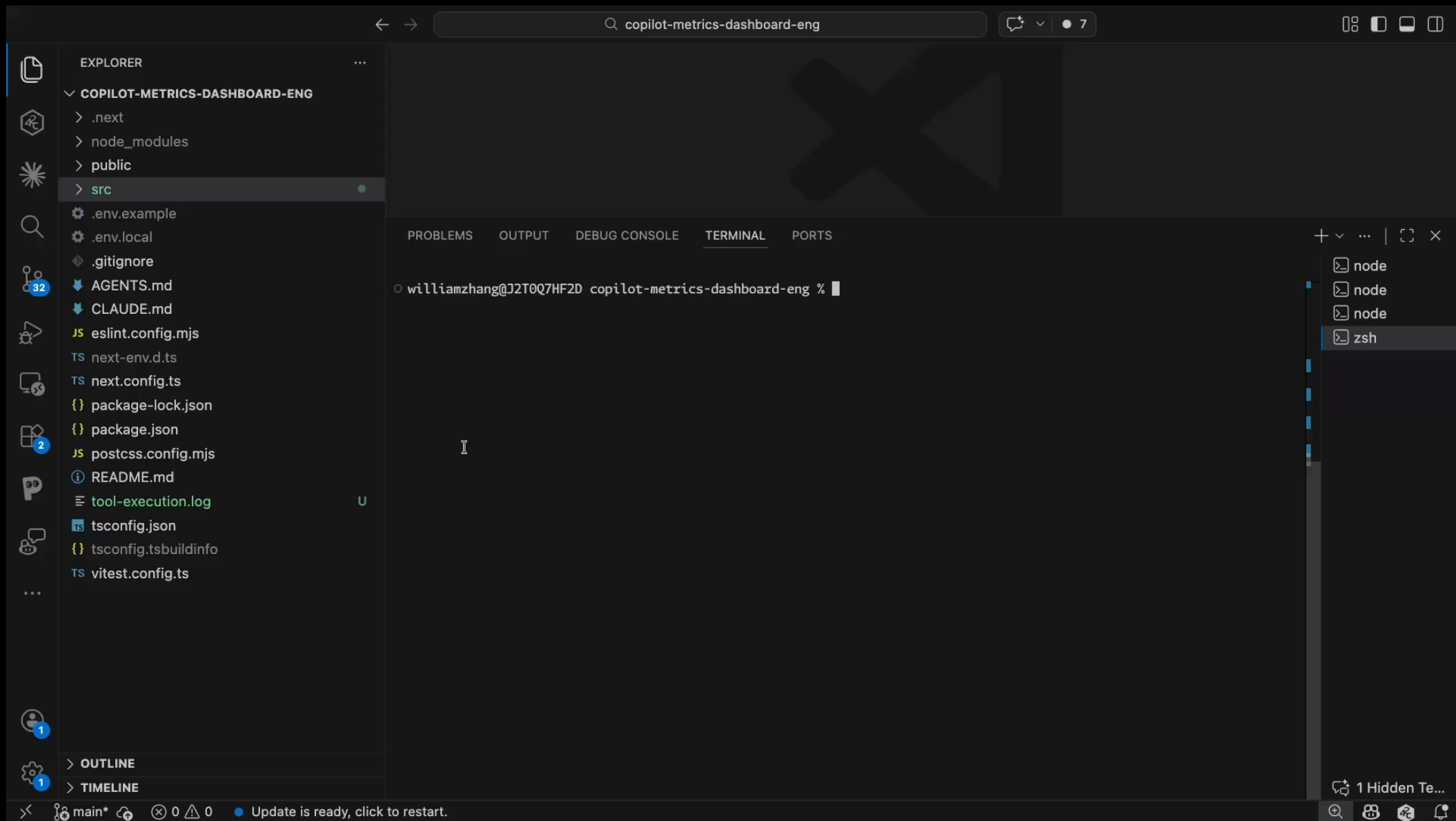
- Natural language → AI determines what to process
- Flexible triggers: scheduled runs, event triggers, comment commands, and more

Getting Started

- Install the gh aw CLI extension
- Create a Markdown file → Compile → Commit
- See Peli's Agent Factory for sample use cases



Demo





--- GitHub Copilot CLI ---

Mac	<code>brew install copilot-cli</code>
Windows	<code>winget install GitHub.Copilot</code>
Linux	<code>npm install -g @github/copilot</code>

--- More Details---

GitHub Copilot CLI	https://docs.github.com/ja/copilot/how-tos/copilot-cli/set-up-copilot-cli/install-copilot-cli
--------------------	---

GitHub Copilot SDK	https://github.com/github/copilot-sdk
--------------------	---

GitHub Agentic Workflow	https://github.github.com/gh-aw/setup/creating-workflows/
-------------------------	---

GitHub Copilot App



GitHub Copilot App (Technical Preview)



GitHub Internal AI-Driven Development

~GitHub Copilot CLI~



Copilot CLI is growing quickly

174

Active
Pull Requests

106

Merged
Pull Requests

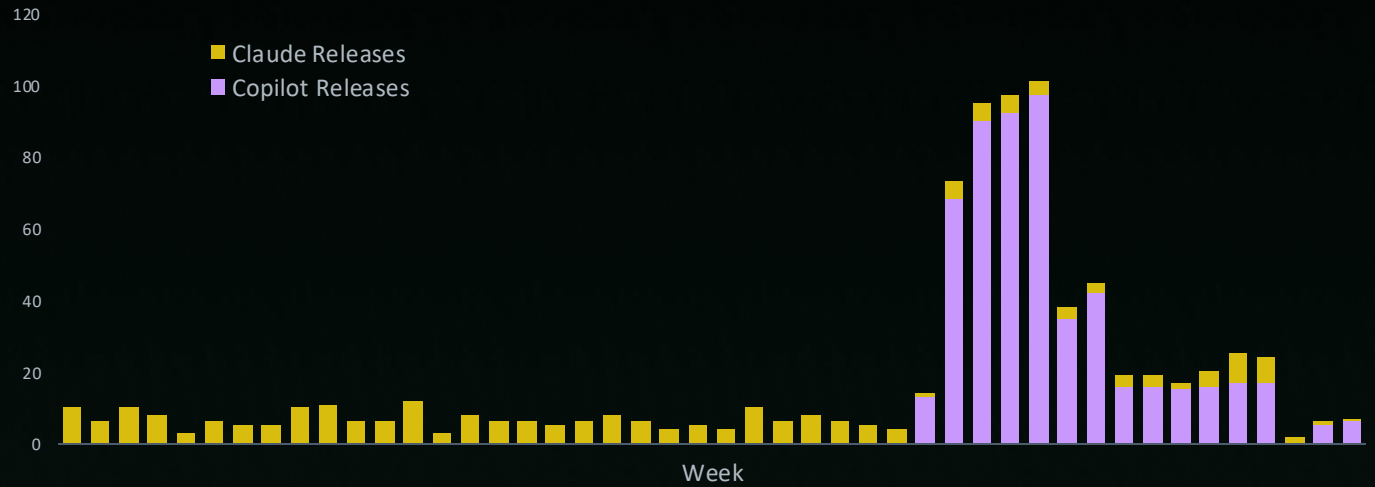
68

Open
Pull Requests

9,678

Commits

Weekly Releases by Product
Copilot CLI vs Claude



Copilot CLI (sonnet-4.5)

Claude Code (sonnet-4.5)

SWE Bench

352/500 (70%)

358/500 (71.6%)

Terminal Bench

48/95 (50.3%)

48/95 (50.5%)

GitHub Internal AI-Driven Development (Example)

Background

Results: 500 PRs Merged/Week

Team Projects:

- Copilot Cloud Agent Runtime
- Copilot Code Review Runtime
- GitHub Copilot CLI
- GitHub Copilot SDK

A single repository has 70K lines of code, of which 54% are tests

Lessons Learned

- Strengthen the feedback loop
- Maintain guardrails
- PR quality management

• Using tracked text files only and excluding build/vendor output like `dist/`, `coverage/`, and `node_modules/`:

Bucket	Files	Physical lines	Non-empty lines
Tests	1,479	398,180	338,042
Prod code	719	20,046	20,046
Benchmarks	257	33,037	29,294
Config/misc	75	20,860	20,600
Docs	62	11,010	7,944
Generated/types	6	10,110	10,088
Scripts/ops	61	8,530	7,312
CI/workflows	47	6,341	5,523

Total: 739,968 physical lines, 644,996 non-empty lines.

Big picture: tests are ~54% of the repo; prod code is ~34%.

PR Velocity Summary

Metric	Last 7 Days (Mar 18-24)	Previous 7 Days (Mar 11-17)	Change
Total merged PRs	503	491	+2.4%
Avg PRs/day	71.9	70.1	+2.4%
Weekday merges	458	453	+1.1%
Weekend merges	45	38	+18.4%

Daily Breakdown

2026-03-24 (Tue): 61 #####
2026-03-23 (Mon): 55 #####
2026-03-22 (Sun): 21 ####
2026-03-21 (Sat): 24 #####
2026-03-20 (Fri): 116 #####
2026-03-19 (Thu): 106 #####
2026-03-18 (Wed): 120 ##### ← Peak day

2026-03-17 (Tue): 80 #####
2026-03-16 (Mon): 86 #####
2026-03-15 (Sun): 24 ####
2026-03-14 (Sat): 14 ###
2026-03-13 (Fri): 80 #####
2026-03-12 (Thu): 117 #####
2026-03-11 (Wed): 90 #####

events.goldcast.io is sharing your screen. Stop sharing Hide

GitHub Internal AI-Driven Development (Example)

What We Couldn't Achieve

SNS development practices trending on social media to GitHub's development could not be adopted

- Multi-Agent Orchestration
 - Fully Automated Development (Ralph Wiggum Loop)
- Patterns that succeed in solo development are hard to scale across teams and organizations
- We often hear that many companies who actually tried it ended up reverting to a stabilization phase

1. Multi-Agent Orchestration

Gas Town

Multi-agent orchestration system for Claude Code with persistent work tracking

Overview

Gas Town is a workspace manager that lets you coordinate multiple Claude Code agents working on different tasks. Instead of losing context when agents restart, Gas Town persists work state in git-backed hooks, enabling reliable multi-agent workflows.

What Problem Does This Solve?

Challenge	Gas Town Solution
Agents lose context on restart	Work persists in git-backed hooks
Manual agent coordination	Built-in mailboxes, identities, and handoffs
4-10 agents become chaotic	Scale comfortably to 20-30 agents
Work state lost in agent memory	Work state stored in Beads ledger

Architecture

2. Fully Automated Development

Ralph Wiggum Plugin

Implementation of the Ralph Wiggum technique for iterative, self-referential AI development loops in Claude Code.

What is Ralph?

Ralph is a development methodology based on continuous AI agent loops. As Geoffrey Huntley describes it: **"Ralph is a Bash loop"** - a simple `while true` that repeatedly feeds an AI agent a prompt file, allowing it to iteratively improve its work until completion.

The technique is named after Ralph Wiggum from The Simpsons, embodying the philosophy of persistent iteration despite setbacks.

Core Concept

This plugin implements Ralph using a **Stop hook** that intercepts Claude's exit attempts:

```
# You run ONCE:
/ralph-loop "Your task description" --completion-promise "DONE"

# Then Claude Code automatically:
# 1. Works on the task
# 2. Tries to exit
# 3. Stop hook blocks exit
# 4. Stop hook feeds the SAME prompt back
# 5. Repeat until completion
```

The loop happens **inside your current session** - you don't need external bash loops. The Stop hook in `hooks/stop-hook.sh` creates the self-referential feedback loop by blocking normal session exit.

This creates a **self-referential feedback loop** where:

- The prompt never changes between iterations
- Claude's previous work persists in files
- Each iteration sees modified files and git history
- Claude autonomously improves by reading its own past work in files

GitHub Internal AI-Driven Development (Example)

What We Did: Improving Agent Experience

1. Faster Tests
2. Stricter Linters
3. Deterministic Formatting
4. Improved CI Feedback Loop
5. Smaller, More Verifiable Units of Work

E2E Implemented test response caching, reducing run time from 2 minutes to 25 seconds

E2E When it grew further, we sharded into 4 parallel executions, 12 from 12 minutes to 4 minutes

E2E test response caching #884

SteveSandersonMS merged 9 commits into [main](#) from [stevesa/e2e-caching](#) on Dec 12, 2025

Conversation 13 · Commits 9 · Checks 55 · Files changed 37

SteveSandersonMS commented on Dec 10, 2025 · edited

Overview

What

Updates the E2E tests (currently CLI only, but that will change soon) so they replay CAPI responses from snapshots.

Why

I'm doing this now as a lead-up to #681 for Copilot SDK. The same mechanism build here will be usable for that too.

Benefits/design

- Makes the E2E tests run a lot faster. In CI it halves the run duration, and locally it has a bigger impact still (for me, reduces time from ~12m to ~3-4m)
 - Note there are a few caveats
- Means we can scale innately and non-deterministically (e.g. if you make a PR and then a second one, the second one can be run in parallel with the first one)
- Makes it super obvious what's happening
 - The snapshots are happening in a separate directory
 - If you make a PR and then a second one, the second one can be run in parallel with the first one
 - This already exists in the CI
- Language-agnostic
 - The intent is to use Node, etc.) as v1 will be shareable
 - This will not require HTTP-level protocol (e.g. COPILOT_API_UI expected snapshots any language-specific)

Core mechanism

- For replay to be useful for-byte HTTP level headers, different test runners need to be able to replay the same headers.

Chore: Add test sharding to runtime-test and tools-test jobs for faster CI #1615

tiagonbotoelho merged 8 commits into [main](#) from [copilot/investigate-build-time-issues](#) 2 days ago

Conversation 11 · Commits 8 · Checks 43 · Files changed 2

Copilot AI commented 4 days ago · edited

Overview

What

Implements test sharding for both `runtime-test` and `tools-test` jobs using Vite's `--shard` flag to significantly reduce CI execution time.

Runtime Test Sharding:

- 4-way sharding reduces execution from ~12m to ~3-4m
- `runtime-test-shard`: Matrix job running 4 parallel shards
- `runtime-test`: Roll-up job that aggregates results and reports the expected status name

Tools Test Sharding:

- 3-way sharding on each OS (macOS, Windows) = 6 parallel jobs total
- `tools-test-shard`: Matrix job running 3 shards x 2 OSes
- `tools-test`: Roll-up job that aggregates results and reports the expected status name

Also updates `download-coverage-reports.sh` to collect all shard-specific coverage artifacts.

Why

`runtime-test` and `tools-test` were the CI bottlenecks at 12+ minutes and 11+ minutes respectively. Branch protection requires jobs named exactly `runtime-test` and `tools-test`, which matrix jobs don't provide (`runtime-test (1)`, etc.), hence the roll-up pattern for both.

Reviewers

Copilot

devm33

tiagonbotoelho

Assignees

tiagonbotoelho

Copilot

Labels

skip-changelog

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Ensure the flows of messages, tool calls, etc are captured in tests #681

[Edit](#)[New issue](#)[Closed](#)

SteveSandersonMS opened on Dec 1, 2025 · edited by SteveSandersonMS

Edits ...

Review the vocab about turns too: we want "turn" to end only when the agent becomes quiescent (currently that's `session.idle`).

Also need to be able to continue a session with follow-up user messages.

In progress: [#943](#)

Create sub-issue



SteveSandersonMS added this to the [SDK 0: Working Preview](#) milestone on Dec 1, 2025



SteveSandersonMS added this to [CCA SDK migration](#) on Dec 1, 2025



github-project-automation moved this to Todo in [CCA SDK migration](#) on Dec 1, 2025



SteveSandersonMS mentioned this on Dec 2, 2025

[CCA v2: Fix "response" event and thereby fix generating PR descriptions #695](#)

[\[Epic\] Copilot SDK #438](#)

Assignees



SteveSandersonMS

Assign to Copilot

Labels



No labels

Type



No type

Projects



1 closed project

Milestone



No milestone

Relationships



None yet

Development



Cloud Agent – For implementation areas with frequent bugs, reinforced test coverage

Move `/runtime` to a separate repository. #8107

Edit <> Code

Merged jmosley merged 7 commits into `main` from `jmosley/cutover-runtime` on Nov 3, 2025

Conversation 2

Commits 7

Checks 42

Files changed 300+

+748 -232,942



jmosley commented on Nov 2, 2025 · edited

...

Overview

What

- Change the `/runtime` to a submodule referencing <https://github.com/github/copilot-agent-runtime>
- Remove the `/benchmarks` directory, and change references to point to `/runtime/benchmarks`.

Something that needs to be resolved: There are tests and build steps for the runtime that require `internal/onboarding/onboarding_prompt.md`. To solve this, I duplicated this file into `copilot-agent-runtime`, but this will lead to a diversion of these two files, and we need a solution to DRY it up.

Why

[github/copilot-agent-runtime#2](https://github.com/github/copilot-agent-runtime#2)

See this gist for a report on the diffs that had to be made during the transition:

<https://gist.github.com/jmosley/9eeb2ff0f58957da626bf4cb66c5d043>



jmosley added 5 commits 2 months ago



Replace runtime with a submodule, and reference benchmarks directory ...

175a7f5



Move docs

f543dde

Reviewers



 Copilot



 tiagonbotelho



 copilot-cli



 coding-agent-runtime-reviewers

Assignees

No one—[assign yourself](#)

Labels

done

Projects



 Copilot coding agent

Status: Done

+19 more

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

When the repository grew too large, the `/runtime` repository and the Copilot CLI repository was split out

Enable typechecked linting, grandfather all existing errors #35

Edit <> Code

Merged friggeri merged 1 commit into [main](#) from [frg/typechecked-lint](#) on Nov 3, 2025

Conversation 2 Co



friggeri commented on I

Same thing as [github/sw](#)



Enable typechec

Copilot AI revie

friggeri requeste
coding-agent-revie



Copilot AI reviewe

Filters

Clear current search query, filters, and sorts

☐ 0 Open ☒ 5 Closed

Author

Label

Projects

Milestones

Reviews

Assignee

Sort

☐ More long tail lint cleaning ✓

#124 by friggeri was merged on Nov 7, 2025 • Approved 2 of 3 tasks

6

☐ Fix most @typescript-eslint/restrict-template-expressions lint errors ✓

#93 by friggeri was merged on Nov 6, 2025 • Approved 2 of 3 tasks

20

☐ Fix all @typescript-eslint/no-unnecessary-type-assertion lint errors ✓

#91 by friggeri was merged on Nov 5, 2025 • Approved 2 of 3 tasks

4

☐ Fix all @typescript-eslint/restrict-template-expressions lint errors ✗

#86 by friggeri was closed on Nov 5, 2025 • Review required 2 of 3 tasks

5

☐ Enable typechecked linting, grandfather all existing errors ✓

#35 by friggeri was merged on Nov 3, 2025 • Approved

2

ProTip! Find everything you created by searching [author:EvanBoyle](#).

Enabled all linter type-checks on the repository

Actual Pull Request Flow

Coding work time 30minutes



Copilot Code Review check



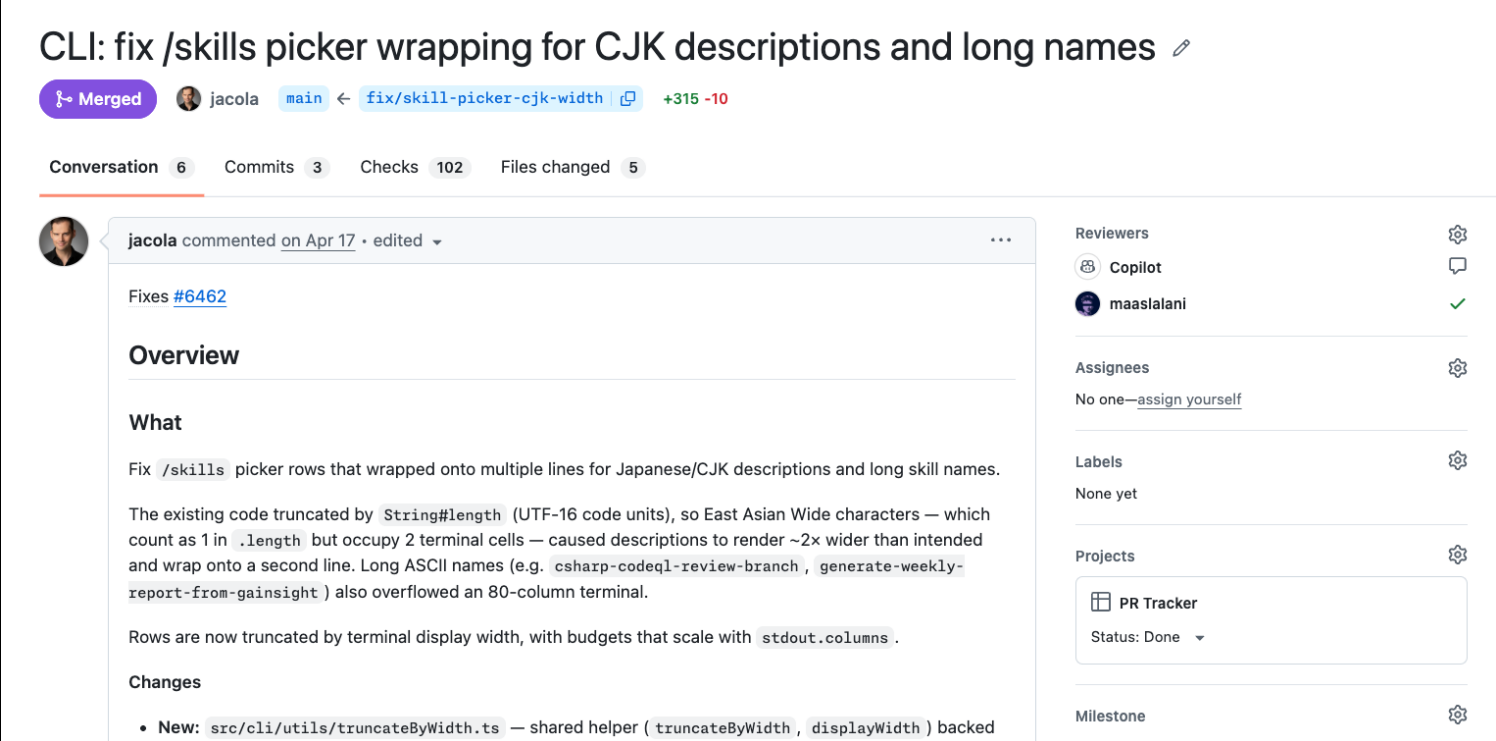
102 GitHub Actions checks 12minutes

- Build & Test (build / lint / test)
- CLI E2E Tests
- Security / Compliance Checks ↓

Request human PR review on Slack 



Ship!! 



The screenshot shows a GitHub Pull Request (PR) titled "CLI: fix /skills picker wrapping for CJK descriptions and long names". The PR is from the "fix/skill-picker-cjk-width" branch to the "main" branch, with a status of "Merged". It shows 6 conversations, 3 commits, 102 checks, and 5 files changed. The PR is reviewed by "maaslalani" with a green checkmark. The "Overview" section includes a "What" section describing the fix: "Fix /skills picker rows that wrapped onto multiple lines for Japanese/CJK descriptions and long skill names. The existing code truncated by String#length (UTF-16 code units), so East Asian Wide characters — which count as 1 in .length but occupy 2 terminal cells — caused descriptions to render ~2x wider than intended and wrap onto a second line. Long ASCII names (e.g. csharp-codeql-review-branch, generate-weekly-report-from-gainsight) also overflowed an 80-column terminal. Rows are now truncated by terminal display width, with budgets that scale with stdout.columns." The "Changes" section lists a new file: "src/cli/utils/truncateByWidth.ts — shared helper (truncateByWidth, displayWidth) backed".

Actual Pull Request Flow

Coding work time 30minutes



Copilot Code Review check



102 GitHub Actions checks 12minutes

- Build & Test (build / lint / test)
- CLI E2E Tests
- Security / Compliance Checks ↓

Request human PR review on

Slack 



Ship!! 

GitHub Copilot

Skills (20/36 enabled)

Toggle skills on/off to control which are available to the AI.

```
[ ] (project ログ、データベースの状態、git履歴、コードを調査して変更を  
jp-debug 加えずに問題を ...  
  
[ ] コード変更を分析し、検証テストを実行し、構造化されたド  
jp-describe-pr(project コメントを作成して ...  
[ ] 技術計画をフェーズごとに検証しながら実装します。承認さ  
jp-implement-pla(project れた実装計画があり、 ...  
  
[ ] 徹底的な調査と更新で既存の実装計画を反復します。新しい  
jp-iterate-pla(project 情報やフィードバック ...  
  
[ ] (project git worktreeを使用して同僚のブランチをレビューするた  
jp-local-review めのローカ ...  
  
[ ] jp-research (project 並行サブエージェントを生成して調査結果を統合すること  
odebase t) で、コードベース全体に ...  
[ ] (project 実装を計画と照合して検証し、成功基準を確認し、問題を特  
jp-validate-plat) 定します。計画を実装 ...  
  
[ ] local-review (project) Set up local environment for review in ...  
[✓] research-codebase (project) Conduct comprehensive research across ...  
[✓] validate-plan (project) Validate implementation against plan, ...  
[✓] weekly-report (project) Searches artifacts (meeting transcrip ...  
[✓] bear (custom) Organize and rename markdown notes an ...  
[✓] create-csp (custom) Update a customer's Success Plan in G...
```


Actual Pull Request Flow

Coding work time 30minutes



Copilot Code Review check



102 GitHub Actions checks 12minutes

- Build & Test (build / lint / test)
- CLI E2E Tests
- Security / Compliance Checks ↓

Request human PR review on

Slack 



Ship!! 



GitHub Internal AI-Driven Development (Example)

Improving the Agent Experience

1. Faster Tests
2. Stricter Linters
3. Deterministic Formatting
4. Improved CI Feedback Loop
5. Smaller, More Verifiable Units of Work

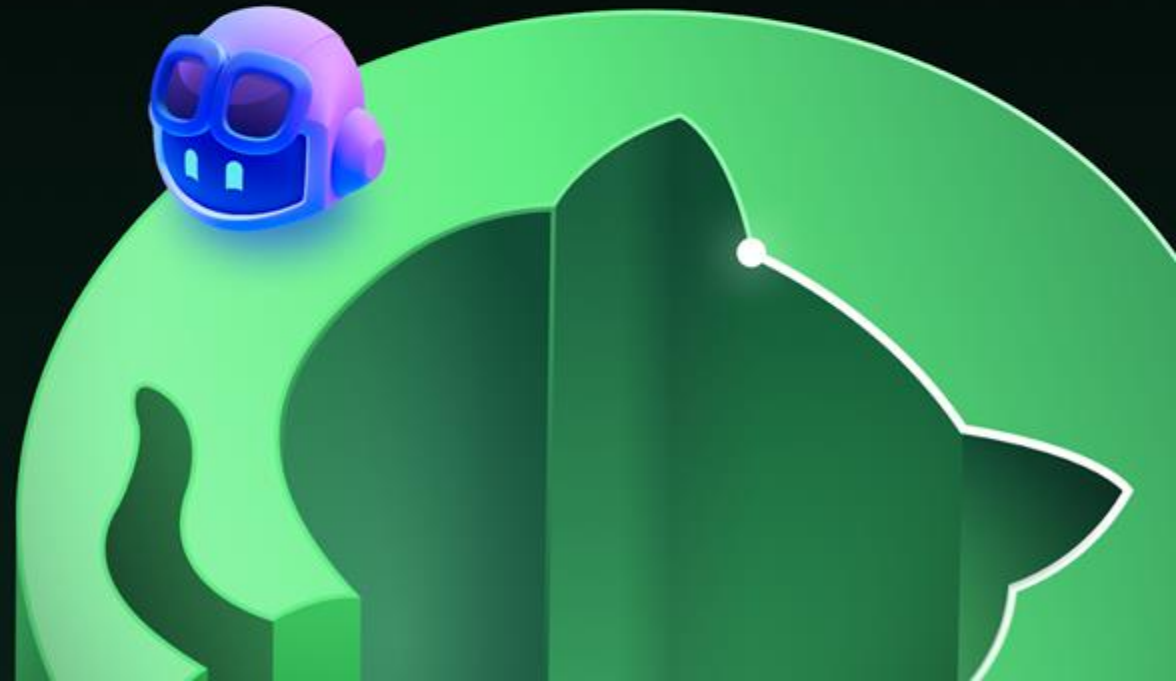
What We Learned

1. The importance of discipline to keep real code, layers, and architecture aligned
2. Emergence of "compounding technical debt"
3. Code Review Growing backlog, improved via tests and automation

Looking Ahead

1. Code is no longer as precious as it used to be, no need to fear throwing it away or rewriting it
2. Good tests become the safety net

GitHub Certifications





Published Certifications



GitHub Foundations

Covers GitHub basics including repositories, branches, pull requests, and collaboration essentials for working on projects.

Level: Beginner
Validity: 24 Months
Duration: 100 Minutes

GitHub Copilot

Tests your ability to use GitHub Copilot to generate code across languages, boost productivity, and streamline development.

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes

GitHub Advanced Security

Focuses on GitHub's security tools, including code scanning, secret scanning, and securing repositories and workflows.

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes

GitHub Actions

Validates skills in creating workflows, automation, and CI/CD pipeline management with GitHub Actions.

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes

GitHub Administration

Demonstrates knowledge of GitHub Enterprise administration, managing users, permissions, integrations, and security.

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes



Published Certifications



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GitHub Administration

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Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes

GitHub Agentic AI Developer

Focused on how developers and teams operate, supervise, and integrate AI agents across the software development lifecycle (SDLC)

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes



Published Certifications



GitHub Foundations

Target Audience:

GitHub users who want to validate their understanding of the foundational topics, products, and concepts of collaborating, contributing, and working on GitHub. Good for **admins/engineers who utilize GitHub platform on daily basis.**

GitHub Foundations

Covers GitHub basics including repositories, branches, pull requests, and collaboration essentials for working on projects.

Level: Beginner
Validity: 24 Months
Duration: 100 Minutes

Study Guide:

https://assets.ctfassets.net/wfutmusr1t3h/1kmMx7AwI4qH8yIZgOmQlP/79e6ff1dfdee589d84a24dd763b1eef7/github-foundations-exam-study-guide__1_.pdf

Resources:

<https://learn.microsoft.com/en-us/training/paths/github-foundations/>

<https://github.com/skills-dev/introduction-to-github>



Published Certifications



GitHub Copilot

Tests your ability to use GitHub Copilot to generate code across languages, boost productivity, and streamline development.

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes

GitHub Copilot

Target Audience:

Individuals who have a foundational understanding of GitHub Copilot with hands-on experience in optimizing software development workflows using GitHub Copilot. Good for **anyone who is interested in Generative AI**.

Study Guide:

<https://assets.ctfassets.net/wfutmusr1t3h/3i7ISEUsTLBgOGrWrML07y/dd586e2b2b607988e2679ed8cce36a76/github-copilot-exam-preparation-study-guide.pdf>

Resources:

<https://www.youtube.com/watch?v=n0NlxUyA7FI>

<https://learn.microsoft.com/en-us/training/paths/copilot/>

<https://github.com/skills/getting-started-with-github-copilot>



Published Certifications

GitHub Advanced Security

Target Audience:

For individuals who have a deep understanding of GitHub and its security features. Good for **anyone who is interested in Application security** such as **SAST, SCM, and Secret Protection**.



GitHub Advanced Security

Focuses on GitHub's security tools, including code scanning, secret scanning, and securing repositories and workflows.

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes

Study Guide:

https://assets.ctfassets.net/wfutmusr1t3h/4WQrNeENScZlISZKdKnVbK/fa664e4ba61056df9194331de18aec0b/github-advanced-security-exam-preparation-study-guide__4_.pdf

Resources:

<https://learn.microsoft.com/en-us/training/paths/github-advanced-security/>
<https://github.com/skills-dev/introduction-to-codeql>



Published Certifications

GitHub Actions

Target Audience:

DevOps engineers, software developers, and IT professionals with experience in GitHub Actions, including workflow creation, automation, and CI/CD pipeline management. Good for **ones interested in CI/CD pipelines**.

Study Guide:

https://assets.ctfassets.net/wfutmusr1t3h/2mMJ6nECbUAdiQMTObbPw6/67cfbffa68fed774a1d280c6c1346635/github-actions-exam-preparation-study-guide__3_.pdf

Resources:

<https://learn.microsoft.com/en-us/training/paths/github-actions/>
<https://learn.microsoft.com/en-us/training/paths/az-400-implement-ci-azure-pipelines-github-actions/>



GitHub Actions

Validates skills in creating workflows, automation, and CI/CD pipeline management with GitHub Actions.

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes



Published Certifications

GitHub Administration

Target Audience:

Designed for system administrators and IT professionals with experience in GitHub Enterprise Administration. Good for **ones in IT department managing security and policies of GitHub Enterprise.**

Study Guide:

https://assets.ctfassets.net/wfutmusr1t3h/5zTfUfFWQknwoUVA1SAw0o/16e2f7904ae2f5cdeb7d91ad32d4547e/github-administration-exam-preparation-study-guide__2_.pdf

Resources:

<https://learn.microsoft.com/en-us/training/paths/github-administration-products/>

<https://www.linkedin.com/learning/paths/prepare-for-the-github-administration-certification>



GitHub Administration

Demonstrates knowledge of GitHub Enterprise administration, managing users, permissions, integrations, and security.

Level: Intermediate
Validity: 24 Months
Duration: 100 Minutes



Published Certifications

GitHub Agentic AI Developer

Target Audience:

Designed for developers, AI/ML engineers, DevOps engineers, platform engineers, solution architects, and product managers who build, supervise, and integrate AI agents in software development workflows. Good for **anyone responsible for designing, integrating, or operating agentic AI systems across the SDLC.**

Study Guide: <https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/gh-600>

Resources:

<https://learn.microsoft.com/en-us/credentials/certifications/github-certified-agentic-ai-developer-beta/>

<https://techcommunity.microsoft.com/blog/skills-hub-blog/new-github-certified-agentic-ai-developer/4517571>



GitHub Agentic AI Developer

Focused on how developers and teams operate, supervise, and integrate AI agents across the software development lifecycle (SDLC)

Level: Intermediate
Validity: 24 Months
Duration: 120 Minutes



Partner Certifications



GitHub Sales Professional

Equips you with the foundational knowledge to confidently position and sell GitHub Enterprise solutions.

Level: 101
Duration: 2-hours



GitHub Tech Sales Professional

Equips partners with the skills to confidently deliver value-driven GitHub demos tailored to a variety of customer personas.

Level: 201



Thank you